

Report
Of
Crop and Weed Detection

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: 25SCS1003000944
Name of Student: **Yash Purohit**
Batch: 2026-27

2026-27

1. CANDIDATE'S DECLARATION

I, YASH, hereby declare that the internship report titled “Crop and Weed Detection” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All references used in preparing this report have been acknowledged. I affirm that this report is my own work and has not been submitted to any other institute for a degree or diploma requirement.

Signature: _____

Student Name: Yash Purohit

Roll No.: 25SCS1003000944

2. ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful to IILM University and the School of Computer Science and Engineering for providing the academic framework for the internship. I am also grateful to UniConverge Technologies and upSkill Campus for providing the internship opportunity, project materials and practical exposure to Data Science and Machine Learning. I appreciate the guidance, learning resources and feedback that helped me understand the project workflow and complete the required documentation.

I express my gratitude to my parents for their blessings and continuous support.

Signature: _____

Student Name: Yash Purohit

Roll No.: 25SCS1003000944

3. INTERNSHIP COMPLETION CERTIFICATE

A copy of the Internship Completion Certificate issued by the industry/organization is attached at the end of this report. The certificate records the internship period as 11-08-2026 to 17-08-2026, the domain as Data Science And Machine Learning, and identifies UniConverge Technologies & upSkill Campus as the organizations associated with the program. Certificate ID: USC251500TIA.

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4. PROJECT DESCRIPTION

4.1 Introduction

Crop and Weed Detection is a computer-vision and machine-learning project in the agricultural domain. The project focuses on detecting crop and weed objects in images and producing object-detection results using class labels and bounding boxes. The implementation prepared during the internship follows a YOLO-based workflow and is written in Python.

4.2 Organization Profile

The internship was undertaken through the UniConverge Technologies and upSkill Campus internship program. The official completion certificate identifies the internship domain as Data Science And Machine Learning and records UniConverge Technologies & upSkill Campus as the organizations associated with the program.

4.3 Problem Statement

Manual identification of weeds from agricultural images can be time-consuming and inconsistent at scale. The project addresses the problem of automatically locating and distinguishing crop and weed objects in images using an object-detection approach. The studied dataset contains sesame crop and weed images with labels.

4.4 Project Objectives

- Understand the crop and weed detection problem and its application in agriculture.
- Prepare and validate the expected dataset structure.
- Generate a YOLO-compatible dataset configuration file.
- Understand preprocessing, augmentation and bounding-box annotation.
- Provide a Python workflow for YOLO model training and image prediction.
- Document implementation, limitations and future scope.

4.5 Scope of the Project

The project scope covers dataset organization, validation of training and validation folders, creation of the YOLO dataset configuration, model-training workflow and prediction on an input image. The submitted code is a baseline implementation and does not claim a measured final accuracy. Future work can include larger datasets, systematic evaluation and field-oriented deployment.

4.6 Technologies and Tools Used

Python is the primary programming language. The implementation uses the Ultralytics YOLO framework for object detection and PyYAML for creating the dataset configuration. Key concepts include computer vision, object detection, image preprocessing, data augmentation, bounding-box annotation, dataset configuration, model training and prediction.

4.7 System Architecture

Agricultural Images → Dataset Preparation and Validation → YOLO Dataset Configuration → YOLO Object-Detection Model → Training → Trained Weights → Image Prediction → Bounding-Box Results. The program creates dataset.yaml, validates the required folders, and then provides training or prediction options.

4.8 Methodology

1. Arrange images and YOLO-format labels in training and validation folders.
2. Create dataset.yaml with dataset paths and classes “crop” and “weed”.
3. Validate required image and label folders.
4. Train YOLO11n using 50 epochs, 512 image size and batch size 8 when a suitable dataset is available.
5. Load the best.pt weights and predict on an input image using a 512 image size and 0.25 confidence threshold.
6. Review the saved detection results.

4.9 Expected Outcomes

The expected outcome is a functional baseline object-detection workflow capable of preparing a YOLO dataset, validating its structure, training a model when a suitable dataset is supplied, and performing prediction. No numerical accuracy, precision, recall or mAP value is claimed because the submitted implementation is a baseline/template without measured evaluation.

4.10 Certificates of completion and other communication mail proof

The official Internship Completion Certificate is attached as an appendix. It confirms successful completion from 11-08-2026 to 17-08-2026 in the Data Science And Machine Learning domain and carries Certificate ID USC251500TIA. Relevant IILM communication/submission material is also attached as supporting proof.

5. BIBLIOGRAPHY / REFERENCES

1. Internship project documentation and learning material provided during the Crop and Weed Detection internship.
2. IILM University, “Internship Report Format updated 26-27”, School of Computer Science and Engineering.
3. Official Internship Completion Certificate issued by UniConverge Technologies & upSkill Campus, Certificate ID USC251500TIA.
4. Ultralytics YOLO documentation and software framework concepts used by the implementation.
5. Python and PyYAML documentation/concepts relevant to the submitted implementation.

INTERNSHIP



CERTIFICATE OF COMPLETION

awarded to

Yash

for successfully completing the Internship Program from **11-08-2026** to **17-08-2026**.

During this internship, the candidate actively participated in hands-on training and real industry project work in the domain of **Data Science And Machine Learning**

This program was conducted in collaboration with **All India Council for Technical Education (AICTE)** and **UniConverge Technologies & upSkill Campus**.



Certificate ID: USC251500TIA

Kaushlendra Singh Sisodia
Chief Mentor, The IoT Academy
Director, UniConverge Technologies

Submission of Summer Internship data for the AY 2026-27

Apoorva Jain <apoorva.jain@iilm.edu>

Wed, Jul 29, 2026 at 3:53 PM

To: CSE 2023-27 GN IILMU <cs27.gn@iilm.edu>, "B. Tech CSE 2024-28" <cs28.gn@iilm.edu>

Cc: "Dr. Lalit Kumar" <lalitkumar@iilm.edu>, "A. Vadivel" <a.vadivel@iilm.edu>, Industry Connect <industryconnect@iilm.edu>, "Mr. ANIK ACHARJEE" <anik.acharjee@iilm.edu>, "Mr. Hilal Ahmad Shah" <hilal.shah@iilm.edu>, "Mr. Ramavath Ganesh" <ramavath.ganesh@iilm.edu>, Surabhi Purwar <surabhi.purwar@iilm.edu>, Vikas Singh <vikas.singh@iilm.edu>, "Ms. Soniya Sharma" <soniya.sharma@iilm.edu>, Director SCSE <director.scse@iilm.edu>

Bcc: "Dr. Manali Gupta" <manali@iilm.edu>, "Dr. Gaurav Kumar" <gauravkumar@iilm.edu>

Dear Students,

As part of your curriculum, you are required to submit the details of the **Summer Internship** you have undertaken.

Please find the link of **Google Form** below and fill in all the required details accurately **on or before 30 August**.

<https://forms.gle/f9Vkfuz3K9YHE4MNA>

Also attached with this email are the **Internship Report Format** and **PPT Guidelines** for your reference.

Internship Report

Kindly prepare your **Internship Report** strictly in the prescribed format.

Internship Presentation (PPT)

There is **no specific format** prescribed for the presentation. However, the **first slide** must include the following:

- IILM University Logo
- IILM University Name
- Internship Topic/Title
- Student Name
- Enrollment Number
- Program & Semester
- Organization Name

You may then proceed with the remaining slides in a professional and well-structured manner.

GitHub Repository Submission

You are required to create a **public GitHub repository** and upload your internship documents to it. Please follow the steps below:

1. Create a **public GitHub repository** using the following naming convention:
Session_Name_RollNo_Semester_Section
2. Upload the following documents to the repository:
 - Internship Report in PDF
 - Internship Presentation (PPT) in PDF
 - Internship Completion Certificate in PDF
3. Ensure that the repository is **Public** so that it is accessible for evaluation.
4. Copy the **GitHub repository link** and submit it in the designated field of the Google Form.

Kindly ensure that all the information provided in the Google Form is accurate and that the GitHub repository contains all the required documents before submitting the repository link.

Faculty Coordinators

For any queries related to the internship submission, please contact your respective faculty coordinator:

Second Year

- **Sections 2CSE1 – 2CSE13:** Ms. Soniya Sharma soniya.sharma@iilm.edu
- **Sections 2CSE14 – 2CSE26:** Mr. Ramvath Ganesh ramavath.ganesh@iilm.edu
- **Sections 2CSE27 – 2CSE39:** Mr. Anik anik.acharjee@iilm.edu

Third Year

- **Sections 3CSE1 – 3CSE13:** Mr. Hilal hilal.shah@iilm.edu
- **Sections 3CSE14 – 3CSE26:** Mr. Vikas Singh vikas.singh@iilm.edu

Final Year

- **Sections 4CSE1 – 4CSE15:** Ms. Surabhi Purwar surabhi.purwar@iilm.edu

Last Date for Submission: 30 August 2026.

PI feel free to contact your Faculty Coordinators for smooth submission of Internship Data.

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Thanks and Regards

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Internship Report Format updated 26-27.docx

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